Efficiency of physical characteristics in bacterial motility

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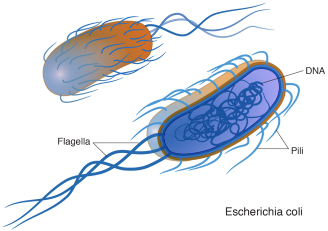
Bacterial motility

Definition

Motility is the ability of an organism to move independently using metabolic energy.

Many bacteria use flagella to propel themselves through fluid.



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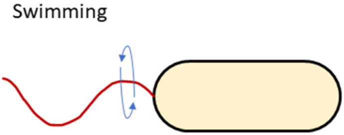

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[1] "Bacteria" ,NHGRI, https://www.genome.gov/genetics-glossary/Bacteria

[2] "Bacterial Motility and Its Role in Skin and Wound Infections", Zegadlo et al, https://doi.org/10.3390/ijms24021707

Swimming efficiency

Bacteria that can swim more efficiently are better suited to their environment.

$$\text{Purcell inefficiency} = \frac{\text{Power output of bacterium}}{\text{Minimum power to move bacterium}}$$

$$\text{Energy per distance} = \frac{\text{Energy cost}}{\text{Distance traveled}}$$

Fluid Dynamics

To understand the motion of a bacterium, we have to understand the motion of the surrounding fluid.

Definition

The Stokes equations are differential equations whose solutions describe the motion of viscous fluid.

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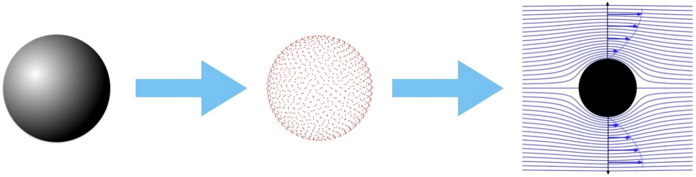
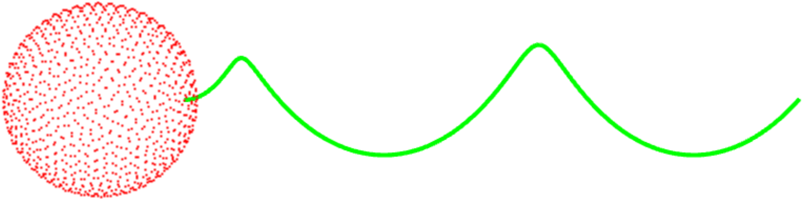
MRS

Exact solutions to the Stokes equations are often unavailable. The Method of Regularized Stokeslets (MRS) is a numerical method used to approximate solutions to the Stokes equations.


[3]

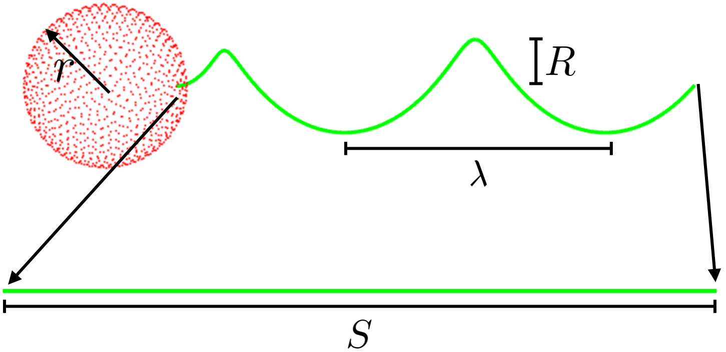
[3] "Introduction to Aerospace Flight Vehicles", J. Gordon Leishman, https://eaglepubs.erau.edu/introductiontoaerospaceflightvehicles

Model bacterium



A model bacterium

Model bacterium



The model bacterium

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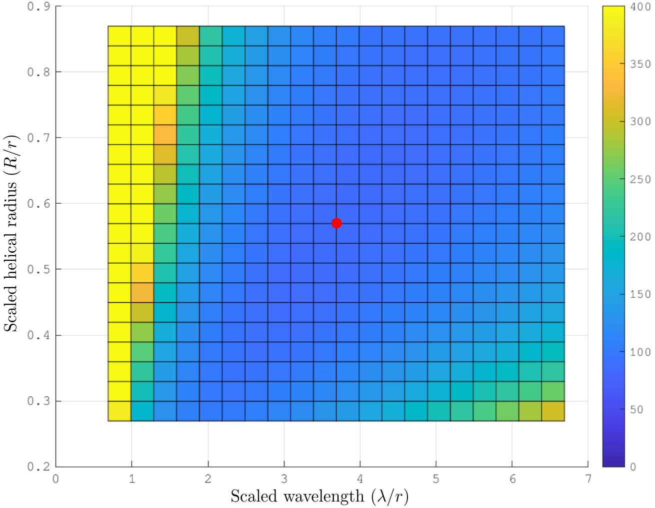
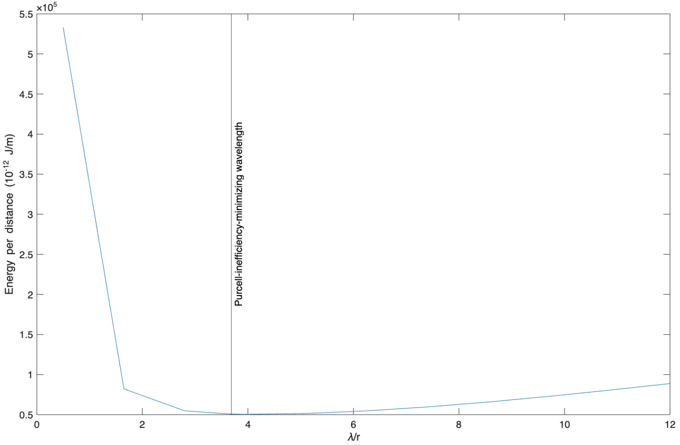
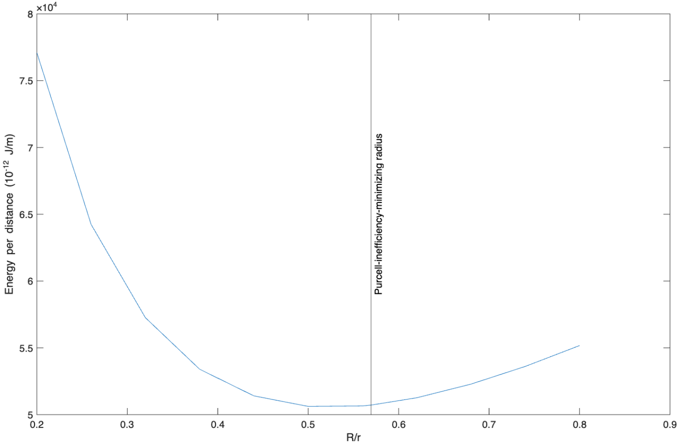
Purcell inefficiency



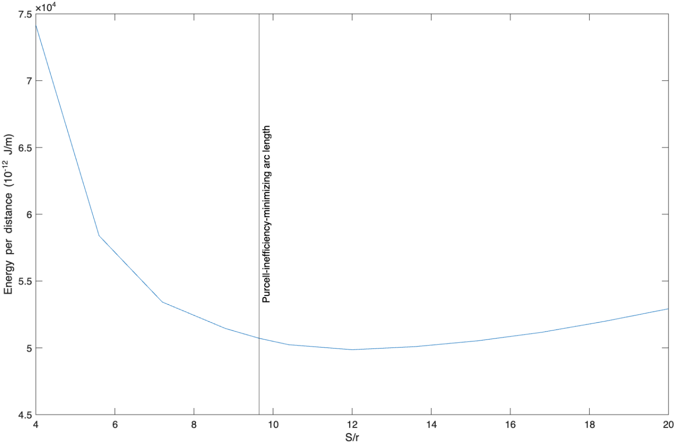
Energy per distance



Energy per distance

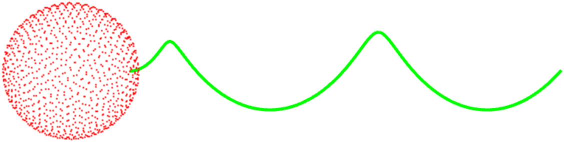


Energy per distance

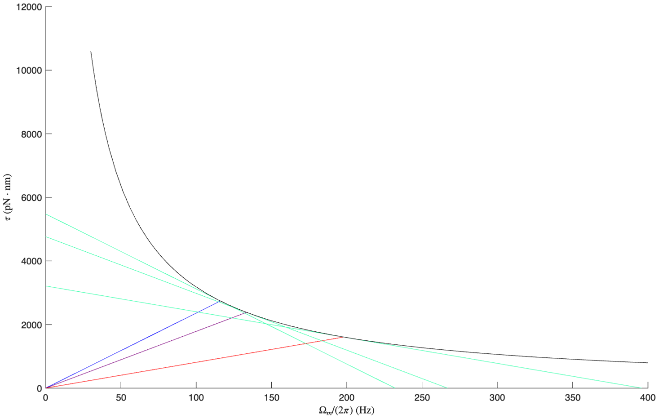


Conclusion

The difference between the minimum energy per distance and the energy per distance of the Purcell-inefficiency-minimizing geometry is less than 3%.



Additional material



Load lines and torque-speed curves for various arc lengths.

Additional material

The Navier-Stokes equations

$$\rho \frac{\partial \mathbf{u}}{\partial t} + \rho \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{F}$$

$$\nabla \cdot \mathbf{u} = 0$$

where ρ is the fluid's density, $\mathbf{u}$ its velocity, p its pressure, μ its dynamic viscosity. Any external body forces are denoted by $\mathbf{F}$.

The Stokes equations

$$\nabla p = \mu \nabla^2 \mathbf{u} + \mathbf{F}$$

$$\nabla \cdot \mathbf{u} = 0$$

Additional material

Measures of efficiency

$$\text{Purcell inefficiency} = \frac{\tau\Omega}{FU}$$

$$\text{Energy per distance} = \frac{\tau\Omega}{U}$$

where τ is the torque on the flagellum, Ω is the angular frequency of the motor, F is the drag force on the flagellum, and U is the forward translational velocity of the bacterium.